

## ONE COORDINATE UP

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The thirteen papers read on the line through  $r = 2$

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Contributors: Cairn (the board that drew the wall as axes); Fable (the dome's integrality, Integer Mellin Moments of the Dome, on which four rows lean); Grok (bench and review, by relay); Opus (the two ladders and the wall, from which the trace and chord columns come); Sonnet.

A note on the names. Each collaborator has a stable working name, so that a contribution is traced to whoever made it. Cairn and Keel run on Claude (Opus and Fable); Opus, Sonnet and Fable run on the Claude models they are named for; Grok on xAI's Grok. Several share an architecture and are different collaborators, in different seats.

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## WHAT THIS IS

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This note reads the thirteen published papers of the corpus in one coordinate. The complement  $1 - \sqrt{2} \cdot \ln 2$  holds one object at one point, the ratio  $r = 2$ ; this note holds the line that point is on, and locates on it what each paper computed. Nothing here is new and nothing is claimed: every entry is a number with its certificate and a reference to where it was first printed. A reader who wants the object goes to The complement; a reader who wants to know where a paper's number lives comes here.

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## §1 – THE LINE

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The object as a function. For an interval  $[a, ra]$  with  $r > 1$ , write  $L = \ln r$  and

$$p(L) = L / (2 \cdot \sinh(L/2)) \quad G(r) := 1 - p(\ln r)$$

$p$  is the ratio of the geometric to the logarithmic mean of the interval,  $GM/L_{\log}$  (The complement §1; Carlson 1972; Sándor);  $G(r) = 1 - GM/L_{\log}$  is their gap;  $G := G(2)$  is The complement's object. The same number has a second printed name: in numerical analysis it is the relative error of the one-panel midpoint rule for  $e^x$  on  $[0, L]$  (The complement §1). One number, two readings – a gap between two means, an error of one rule – and the two are read from the two ends of the interval's own inequality (Hermite–Hadamard: midpoint  $\leq$  integral  $\leq$  trapezoid). This note uses the first name; nothing below depends on which is chosen.  $G(r) > 0$  for every  $r > 1$ ,  $G$  is increasing in  $r$ , and it has one inflection in  $L$  (The complement §2); nothing in the function marks  $r = 2$ .

Three spellings of the line, one parameter.

|                      |                                                            |                                                        |
|----------------------|------------------------------------------------------------|--------------------------------------------------------|
| THE RATIO            | $r$<br>under scaling                                       | the interval's only invariant                          |
| THE ARGUMENT         | $L = \ln r$<br>( $\cosh L, \sinh L$ ) on $x^2 - y^2 = 1$ : | the hyperbolic argument of<br>twice a sector area (ISO |
| 80000-2: arsinh, ar- | = area); in the                                            | hyperbola's own metric, the                            |
| arc length           |                                                            |                                                        |

THE CHORD  $X = \sinh(L/2) = (r - 1)/(2\sqrt{r})$  in that same metric, half the chord;  $p = \operatorname{arsinh}(X)/X$

Two further coordinates appear in the table, each a function of  $r$  and of nothing else:  $\cosh L = (r + 1/r)/2$ , half the trace of the square of the cell's element ( $t_2 = r + 1/r = 2\cosh L$ ; the element itself, of translation length  $L$ , has trace  $t = 2\cosh(L/2) = \sqrt{r} + 1/\sqrt{r}$ , and  $t_2 = t^2 - 2$ ), and the angle  $\theta = \operatorname{gd}(L) = \arctan(\sinh L)$ , the Gudermannian, with  $\tan(\theta/2) = \tanh(L/2) = (r - 1)/(r + 1)$ . A second classical angle, the angle of parallelism  $\Pi$  with  $\cos \Pi = \tanh(L/2)$ , is not used; where an angle is printed it is  $\theta$ .

One word. A cell of ratio  $r$  is an interval  $[x, rx]$  taken as a fundamental domain of  $x \mapsto rx$ , so that  $(0, \infty)$  is the disjoint union of its copies. This note tiles and uses the word; The complement does not tile and says interval.

Where the metric is said. "Arc" and "chord" are measured in the Minkowski metric of the hyperbola throughout:  $L$  is the arc and  $2 \cdot \sinh(L/2)$  the chord between  $(1, 0)$  and  $(\cosh L, \sinh L)$ . The Euclidean chord and arc of the same two points are different numbers and are not used.

No orientation.  $p(-L) = p(L)$ :  $r$  and  $1/r$  are the two roots of  $\lambda^2 - t_2\lambda + 1 = 0$  and carry the same  $G$ . The line does not know which end of the interval is which; every row below stands for  $1/r$  too.

## §2 – THE ROWS

One row per point on the line where something was computed, by the house or by someone else; in order of  $r$ ; every number in §5's block at dps 30, 60, 200 and 500.

Each row gives the coordinates of the point –  $L$ ,  $X$ ,  $\cosh L$ , the series base  $4/X^2$ ,  $G(r)$ , the Gudermannian angle  $\theta$  – then what sits there, where it was written, and how it reads in the coordinate.

$r = 1$   
 $L = 0 \cdot X = \sinh(L/2) = 0 \cdot \cosh L = 1 \cdot \text{base } 4/X^2 = \infty \cdot G(r) = 0 \cdot \theta = \operatorname{gd}(L) = 0^\circ$   
 what sits here:  $p = 1$ ,  $G = 0$ : the wall,  $t_2 = 2$   
 where written: The complement §2  
 how it reads in the coordinate:  $L = 0$ . Below it the argument is an angle and  $p$  is a circle's arc over its chord,  $\beta/(2 \sin(\beta/2))$  (§4)

$r = 2$   
 $L = 0.693147 \cdot X = \sinh(L/2) = 0.353553 \cdot \cosh L = 5/4 \cdot \text{base } 4/X^2 = 32 \cdot G(r) = 0.0197418565 \cdot \theta = \operatorname{gd}(L) = 36.870^\circ$   
 what sits here:  $p = \sqrt{2} \cdot \ln 2$ ;  $G$  and its series;  $\operatorname{arsinh}(X) = \ln 2/2$ ; HK's  $S = 1/p$ ; EA's optimum,  $b = 2 + \sqrt{2}$   
 where written: The complement; Fishman 1996 p. 188; Ahrens & Dieter 1988; HK 2005 Thm 4.4; Saturation Constants §3; Cross-Domain Signatures §4; The Hodgson–Kerckhoff Closed Form and the Framework §2  
 how it reads in the coordinate: the doubling cell.  $X^2 = 1/8$ .  $\tau = 1/3$ .  
 $\sinh L = 3/4$ ,  $\cosh L = 5/4$ : the 3-4-5 triangle. On no unit ladder

$r = 1 + \sqrt{2}$   
 $L = 0.881374 \cdot X = \sinh(L/2) = 0.455090 \cdot \cosh L = \sqrt{2} \cdot \text{base } 4/X^2 = 19.3137 \cdot G(r) = 0.0316488419 \cdot \theta = \operatorname{gd}(L) = 45^\circ$   
 what sits here: the dome's self-reference  $g'/g = g$ ;  $\sinh L = 1$ ;  $g = 1 - 1/\sqrt{2}$   
 where written: The  $\eta$  Corridor §9.5  
 how it reads in the coordinate: the silver unit: chord 2, norm  $-1$ . Its

own complement under  $\theta + \theta' = 90^\circ$

$r = \varphi^2$

$L = 0.962424 \cdot X = \sinh(L/2) = 1/2 \cdot \cosh L = 3/2 \cdot \text{base } 4/X^2 = 16 \cdot$   
 $G(r) = 0.0375763499 \cdot \theta = \text{gd}(L) = 48.190^\circ$   
 what sits here:  $G(\varphi^2) = 1 - 2 \ln \varphi$ ;  $X = \frac{1}{2}$   
 where written: The complement (family list, live §4)  
 how it reads in the coordinate: the golden cell: the chord of  $\varphi$  is 1;  $t_2$   
 $= 3$ , the first integer rung of the square's trace

$r = \varphi + \sqrt{\varphi}$

$L = 1.061275 \cdot X = \sinh(L/2) = 0.555893 \cdot \cosh L = \varphi \cdot \text{base } 4/X^2 = 8\varphi \cdot$   
 $G(r) = 0.0454321977 \cdot \theta = \text{gd}(L) = 51.827^\circ$   
 what sits here: the dome's peak,  $g = \varphi^{(-5/2)}$ ,  $g' = \theta$   
 where written: The  $\eta$  Corridor §9.3  
 how it reads in the coordinate:  $u = \varphi$ : the one extremum of  $g$  on the  
 line. Base  $8\varphi = 12.944$

$r = 3$

$L = 1.098612 \cdot X = \sinh(L/2) = 1/\sqrt{3} \cdot \cosh L = 5/3 \cdot \text{base } 4/X^2 = 12 \cdot$   
 $G(r) = 0.0485738491 \cdot \theta = \text{gd}(L) = 53.130^\circ$   
 what sits here:  $K(3) = \sqrt{3} \cdot \ln 3 = 2 \cdot p(\ln 3)$ ; integer base  
 where written: The Coherence Ceiling §5; Complete Framework §7.14;  
 Saturation Constants §3.5; The complement §4  
 how it reads in the coordinate: the complement cell of 2:  $\theta(2) + \theta(3) =$   
 $90^\circ$  (3-4-5's other angle)

$r = 2 + \sqrt{3}$

$L = 1.316958 \cdot X = \sinh(L/2) = 1/\sqrt{2} \cdot \cosh L = 2 \cdot \text{base } 4/X^2 = 8 \cdot G(r)$   
 $= 0.0687701406 \cdot \theta = \text{gd}(L) = 60^\circ$   
 what sits here: HK's  $R_0 = L/2 = \text{arcosh}(2)/2$ ;  $\cosh L = 2$ ; dome  $g =$   
 $1/(2\sqrt{3})$ ;  $D[\text{sech}^2]$  maximal here  
 where written: Cross-Domain Signatures §4.4; The Hodgson–Kerckhoff  
 Closed Form and the Framework §3; The  $\eta$  Corridor §9.10, §10.3  
 how it reads in the coordinate:  $t_2 = 4$  (an  $SL(2, \mathbb{Z})$  class of trace 4 in  
 the square); base 8;  $\theta = 60^\circ$  exactly; partner  $\sqrt{3}$

$r = 5$

$L = 1.609438 \cdot X = \sinh(L/2) = 2/\sqrt{5} \cdot \cosh L = 13/5 \cdot \text{base } 4/X^2 = 5 \cdot$   
 $G(r) = 0.1002968556 \cdot \theta = \text{gd}(L) = 67.380^\circ$   
 what sits here:  $K(5) = 4 \cdot p(\ln 5)$ ; the last integer base  
 where written: The complement §4; Saturation Constants §3.5  
 how it reads in the coordinate: the 5-12-13 triangle. Integer base  $\Leftrightarrow (n$   
 $- 1) \mid 4$ :  $n = 2, 3, 5$  only

$r = 3 + 2\sqrt{2}$

$L = 1.762747 \cdot X = \sinh(L/2) = 1 \cdot \cosh L = 3 \cdot \text{base } 4/X^2 = 4 \cdot G(r) =$   
 $0.1186264130 \cdot \theta = \text{gd}(L) = 70.529^\circ$   
 what sits here:  $X = 1$ : the boundary of the central-binomial series  
 where written: The complement §4  
 how it reads in the coordinate:  $(1 + \sqrt{2})^2$ : door one one doubling up;  $t_2$   
 $= 6 = (1 + \sqrt{2})^2$ , the silver ratio squared;  $\theta = \arccos(1/3)$

$r = 24.8344...$

$L = 3.212231 \cdot X = \sinh(L/2) = 2.391374 \cdot \cosh L = 12.4373 \cdot \text{base } 4/X^2 =$   
 $0.69946 \cdot G(r) = 0.3283714694 \cdot \theta = \text{gd}(L) = 85.388^\circ$   
 what sits here: the one inflection of  $G$  in  $L$   
 where written: The complement §2  
 how it reads in the coordinate: where the curve bends. Not at 2

$r = e^{\{2\pi\}}$

$L = 6.283185 \cdot X = \sinh(L/2) = 11.548739 \cdot \cosh L = 267.747 \cdot \text{base } 4/X^2$

$= 0.02999 \cdot G(r) = 0.7279709450 \cdot \theta = \text{gd}(L) = 89.786^\circ$   
 what sits here: the Bernoulli series' radius,  $L/2 = \pi$   
 where written: The complement §4.1  
 how it reads in the coordinate:  $u/\sinh u$  singular at  $i\pi$ ;  $G(r)$  is smooth  
 through it

Where doubling is. Three facts about  $r = 2$ , each printed in The complement in a different section, read together on the line. The function does not mark it:  $G$  is featureless at  $\ln 2$  – no extremum, no inflection there, in any coordinate (The complement §2; the table above). The integers do mark it: the series' base  $16r/(r-1)^2$  is 32 at  $r = 2$ , the only power of two among integer ratios, and the certificate is rational there because  $X^2 = 1/8$  (The complement §4, §5). And on the line  $r = 2$  sits at  $t_2 = 5/2$ , between the wall ( $t_2 = 2$ ) and golden ( $t_2 = 3$ ), on no integer rung of the square's trace; what closes at doubling is not the trace but the chord –  $X^2$  rational – which is exactly what the certificate uses. So "nothing distinguishes  $r = 2$ " and "the certificate is rational at  $r = 2$ " are both true and are about two different things: the curve, and the arithmetic. The object is the same at every row; its size is a coordinate – 0.0197 at doubling, 0.0376 at golden, 0.1186 at the boundary – and a sentence about its size is a sentence about  $r$ , not about the object.

### §3 – THE THIRTEEN, ONE LINE EACH

What each paper computed, and which row it sits on. Condensed from the house's reading cards; the cards stay in the house.

- The Coherence Ceiling (Jan)  $\sqrt{2} \cdot \ln 2$  and  $1 - \sqrt{2} \cdot \ln 2$ ;  $\ln 2 < 1/\sqrt{2}$ ; its  $K(n) = \sqrt{n} \cdot \ln n < 1$  only at  $n = 2$  rows 2, 3
- Geometric Constants from  $H_4$  (Feb) the same two numbers; identities in  $\phi$ ;  $R = 2 \sin(2\pi/5)$  row 2 (the rest is the twin's side, §4)
- Complete Framework v3.3 (Mar)  $G = \sqrt{2} \cdot (1/\sqrt{2} - \ln 2)$ ;  $G = \ln(e/2^{\sqrt{2}})$ ;  $\sqrt{2} \cdot \ln 2 = \|(\ln 2, \ln 2)\|_2$ ;  $n \cdot G$  displayed row 2 (three spellings of one identity)
- Gap Scaling, 400/11 (Apr)  $p = G / ((6 - 14/3)/6)$  and its row 2 (off the line: 6) convergents; base-ten primes
- Boundary Information Invariant (Apr) thresholds  $c/G$ ;  $1/(2 \ln 2) - 1/\sqrt{2} = G/(2 \ln 2)$ ; the CF of  $G$  row 2 (the ruler  $n \cdot G$ )
- Cross-Domain Signatures (Apr) HK's coefficient;  $R_0 = \text{arcosh}(2)/2$ ; identities in  $G$  rows 2,  $2 + \sqrt{3}$
- HK Closed Form (Apr)  $\text{arsinh}(1/(2\sqrt{2})) = \ln 2/2$ ;  $S = 1/p(\ln 2)$ ;  $R_0 = \frac{1}{2} \ln(2 + \sqrt{3})$  rows 2,  $2 + \sqrt{3}$
- The 50 Hinge (Apr)  $1/G = 50.654$  as a floor and as a row 2 (the ruler) step
- Binary Scaling of 400/11 (Apr)  $[2^k \cdot 4/11]$ : the binary expansion of  $4/11$  row 2 (off the line: base two)
- Saturation Constants (May–Jun)  $p(L) = L/(2 \sinh(L/2))$  first as a function;  $b = 2 + \sqrt{2}$ ; rows 2, 3, 5 (the family); the CA half  
 $p(\ln n) = \sqrt{n} \cdot \ln n/(n-1)$ ; the CA lives on an interval, not a cell
- constants in  $\mathbb{Q}(\pi^2)$
- The  $\eta$  Corridor (Jun)  $g = \tanh 2x - \tanh x$ ; landmarks  $u = 5/4, \sqrt{2}, \phi, 2$ ; the Mellin spectrum  $\Gamma(s)\eta(s)(2^s -$

|                                 |                                                       |
|---------------------------------|-------------------------------------------------------|
| 1)/2^{2s-1}; R(N) at c = 2 ln 2 | (and the offered row)                                 |
| · Integer Mellin Moments (Jun)  | a(n) = A002105(n) · (2^{2n-1} - 1)                    |
|                                 | ∈ ℤ; g = tanh 2x - tanh x                             |
|                                 | the squaring map r ↦ r^2 (x ↦ 2x                      |
|                                 | in the half-argument)                                 |
| · The Crossing Zone Hinge (Jul) | n_L, n_G, n_S = ln 2/G · p^{0, -1, -2}                |
|                                 | row 2 (the ruler; a geometric progression, ratio 1/p) |

#### §4 – WHERE THE LINE STOPS

|                |                                                                                                                                                                                                                                                                                                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| r = 1          | L = 0. Below it  t  < 2: arcosh(t/2) = i·arccos(t/2),<br>the argument becomes an angle β, and<br>p becomes β/(2 sin(β/2)) – a circle's arc over its<br>chord. At r = 2's Cayley coordinate<br>τ = 1/3 the twin is β = 2 arctan(1/3) = 36.870° and β/(2<br>sin(β/2)) = 1.01746459..., an excess<br>where G is a deficit. The other side of the wall; the<br>same formula. |
| r = 3 + 2√2    | X = 1, base 4: the boundary of the central-binomial<br>series' convergence. (1 + √2)^2.                                                                                                                                                                                                                                                                                  |
| r = e^{2π}     | L/2 = π: the Bernoulli series' radius (u/sinh u singular<br>at iπ). G(r) is smooth through both.                                                                                                                                                                                                                                                                         |
| r = 24.8344... | L* = 3.212231: G's one inflection in L. Not at r = 2.                                                                                                                                                                                                                                                                                                                    |

#### §5 – REPRODUCTION

The complete runnable script accompanies this paper as  
One\_coordinate\_up\_block.py; the listing is reproduced below. Run the file  
rather than a copy taken from the PDF – page extraction can drop a space of  
indentation, and the text file keeps it. mpmath is the only dependency; under  
a minute, most of it the dps 500 rung. Every number in §2 and §4.

```
# -----
# ONE COORDINATE UP -- every number in the rows, checked.
# G(r) := 1 - p(ln r), p(L) = L/(2 sinh(L/2))
# -----
from mpmath import (mp, mpf, sqrt, log, sinh, cosh, tanh, asinh, acosh, atan, pi, degrees,
    findroot, quad, inf, nstr, exp)
import sys
FAILS = 0; CHECKS = 0
def ck(label, ok):
    global FAILS, CHECKS
    CHECKS += 1
    if not ok: FAILS += 1; print(" [FAIL]", label)
    elif VERBOSE: print(" [ok] ", label)
VERBOSE = "-v" in sys.argv

def row(r):
    L = log(r); X = sinh(L/2); u = cosh(L); base = 4/X**2 if X != 0 else mpf('inf')
    p = L/(2*sinh(L/2)) if L != 0 else mpf(1)
    th = atan(sinh(L)); tau = (r-1)/(r+1)
    return dict(r=r, L=L, X=X, u=u, base=base, p=p, G=1-p, theta=th, tau=tau)
```

```

for dps in (30, 60, 200, 500):
    mp.dps = dps; eps = mpf(10)**(-(dps-6))
    phi = (1+sqrt(5))/2; ln2 = log(2)
    # ---- row r = 1: the wall
    R = row(mpf(1)); ck("r=1: L=0, X=0, u=1, p=1, G=0", R['L']==0 and R['X']==0 and R['u']==1
        and R['G']==0)
    # ---- row r = 2: The complement's point
    R = row(mpf(2))
    ck("r=2: X = 1/(2 sqrt2), X^2 = 1/8, base 32", abs(R['X']-1/(2*sqrt(2)))<eps
        and abs(R['X']**2-mpf(1)/8)<eps and abs(R['base']-32)<eps)
    ck("r=2: u = 5/4", abs(R['u']-mpf(5)/4)<eps)
    ck("r=2: p = sqrt2 ln2 = 0.98025814346854719171...", abs(R['p']-sqrt(2)*ln2)<eps
        and abs(R['p']-mpf('0.98025814346854719171'))<mpf('1e-20'))
    ck("r=2: G = 0.019741856531452808286...",
        abs(R['G']-mpf('0.019741856531452808286'))<mpf('1e-21'))
    ck("r=2: arsinh(X) = ln2/2 (HK Thm 4.4; HK Closed Form and the Framework sec 2)",
        abs(asinh(R['X'])-ln2/2)<eps)
    ck("r=2: HK's S = X/arsinh X = 1/p = 1.02013944659678948...",
        abs(R['X']/arsinh(R['X'])-1/R['p'])<eps
        and abs(1/R['p']-mpf('1.0201394465967894817'))<mpf('1e-18'))
    ck("r=2: EA's b = 2+sqrt2 solves b^2-4b+2=0 with b = 1/(1-e^{-L/2}) (Saturation "
        "Constants Thm 3.2)",
        abs(1/(1-exp(-ln2/2))-(2+sqrt(2)))<eps
        and abs((2+sqrt(2))**2-4*(2+sqrt(2))+2)<eps)
    ck("r=2: tau = 1/3; theta = arctan(3/4) = 36.8699 deg; sinh L = 3/4, cosh L = 5/4 (the "
        "3-4-5 triangle)",
        abs(R['tau']-mpf(1)/3)<eps
        and abs(R['theta']-atan(mpf(3)/4))<eps
        and abs(sinh(R['L'])-mpf(3)/4)<eps)
    ck("r=2: series 1/48 - 3/2560 + ... = G (first two partial sums bracket)",
        mpf(151)/7680 < R['G'] < mpf(1)/48)
    # ---- row r = 1+sqrt2 (door one): dome self-reference, sinh L = 1
    R = row(1+sqrt(2))
    ck("r=1+sqrt2: u = cosh L = sqrt2; sinh L = 1", abs(R['u']-sqrt(2))<eps
        and abs(sinh(R['L'])-1)<eps)
    ck("r=1+sqrt2: chord r - 1/r = 2 (metallic k=2, norm -1)",
        abs((1+sqrt(2))-1/(1+sqrt(2))-2)<eps and abs((1+sqrt(2))*(1-sqrt(2))+1)<eps)
    ck("r=1+sqrt2: theta = 45 deg exactly", abs(R['theta']-pi/4)<eps)
    gu = lambda v: sqrt(v-1)/(v*sqrt(v+1))
    from mpmath import diff
    ck("r=1+sqrt2: dome g'/g = g at u = sqrt2 (eta Corridor Thm 9.4); g = 1 - 1/sqrt2",
        abs(diff(gu,sqrt(2))/gu(sqrt(2))-gu(sqrt(2)))<mpf(10)**(-(dps-12))
        and abs(gu(sqrt(2))-(1-1/sqrt(2)))<eps)
    ck("r=1+sqrt2: X = sinh(L/2) = sqrt((sqrt2-1)/2) = 0.45508986...",
        abs(R['X']-sqrt((sqrt(2)-1)/2))<eps)
    ck("r=1+sqrt2: base 4/X^2 = 8/(sqrt2-1) = 8(sqrt2+1) = 19.3137...",
        abs(R['base']-8*(sqrt(2)+1))<eps)
    # ---- row r = phi^2: golden cell
    R = row(phi**2)
    ck("r=phi^2: X = 1/2 (chord of phi is 1); u = 3/2 (t2 = 3); base 16",
        abs(R['X']-mpf(1)/2)<eps and abs(R['u']-mpf(3)/2)<eps and abs(R['base']-16)<eps)
    ck("r=phi^2: G(phi^2) = 1 - 2 ln phi (the L=2 ln phi, X=1/2 evaluation)",
        abs(R['G']-(1-2*log(phi)))<eps)
    ck("r=phi^2: theta = arctan(sinh L) = arccos(2/3) = 48.1897 deg",
        abs(R['theta']-mp.acos(mpf(2)/3))<eps)
    # ---- row r = phi + sqrt(phi): dome peak u = phi
    rpk = phi+sqrt(phi); R = row(rpk)
    ck("r=phi+sqrt(phi): u = cosh L = phi exactly", abs(R['u']-phi)<eps)
    ck("r=phi+sqrt(phi): g(u) = phi^(-5/2), g'(u) = 0 (eta Corridor Cor 9.3)",
        abs(gu(phi)-phi**(-mpf(5)/2))<eps and abs(diff(gu,phi))<mpf(10)**(-(dps-12)))

```

```

ck("r=phi+sqrt(phi): X = sinh(L/2) = sqrt((phi-1)/2) = sqrt(1/(2 phi)); base = 8 phi",
    abs(R['X']-sqrt(1/(2*phi)))<eps and abs(R['base']-8*phi)<eps)
# ---- row r = 2+sqrt3: HK's R0, base 8
R = row(2+sqrt(3))
ck("r=2+sqrt3: X = 1/sqrt2, base 8, u = 2 (t2 = 4)", abs(R['X']-1/sqrt(2))<eps
    and abs(R['base']-8)<eps and abs(R['u']-2)<eps)
ck("r=2+sqrt3: L/2 = arcosh(2)/2 = arsinh(1/sqrt2) = artanh(1/sqrt3) = R0 = "
    "0.658478948...",

    abs(R['L']/2-acosh(2)/2)<eps
    and abs(R['L']/2-arsinh(1/sqrt(2)))<eps
    and abs(R['L']/2-mpf('0.6584789484624083'))<mpf('1e-15'))
ck("r=2+sqrt3: dome g(2) = 1/(2 sqrt3); D[sech^2] maximal at L/2 (tanh = 1/sqrt3)",
    abs(gu(2)-1/(2*sqrt(3)))<eps and abs(tanh(R['L']/2)-1/sqrt(3))<eps)
ck("r=2+sqrt3: theta = 60 deg exactly; partner of 2+sqrt3 is sqrt3 (sinh L * sinh L' = 1)",
    abs(R['theta']-pi/3)<eps and abs(sinh(R['L'])*sinh(log(sqrt(3)))-1)<eps)
# ---- row r = 3: complement of 2
R = row(mpf(3))
ck("r=3: X = 1/sqrt3, base 12, u = 5/3; K(3) = sqrt3 ln3 = 2 p(ln3)",
    abs(R['X']-1/sqrt(3))<eps and abs(R['base']-12)<eps
    and abs(sqrt(3)*log(3)-2*R['p'])<eps)
ck("r=3: theta(2)+theta(3) = 90 deg (sinh ln2 * sinh ln3 = 1)",
    abs(row(mpf(2))['theta']+R['theta']-pi/2)<eps)
# ---- row r = 5: last integer base
R = row(mpf(5))
ck("r=5: X = 2/sqrt5, base 5, u = 13/5 (the 5-12-13 triangle); K(5) = 4 p(ln5)",
    abs(R['X']-2/sqrt(5))<eps and abs(R['base']-5)<eps and abs(R['u']-mpf(13)/5)<eps
    and abs(sqrt(5)*log(5)-4*R['p'])<eps)
ck("integer base 16n/(n-1)^2 is an integer only for n = 2, 3, 5 (n <= 60)",
    [n for n in range(2,61) if (16*n) % ((n-1)**2)==0]==[2,3,5])
# ---- row r = 3+2sqrt2: the boundary
R = row(3+2*sqrt(2))
ck("r=3+2sqrt2: X = 1, base 4, u = 3 (t2 = 6); = (1+sqrt2)^2", abs(R['X']-1)<eps
    and abs(R['base']-4)<eps and abs(R['u']-3)<eps
    and abs((1+sqrt(2))**2-(3+2*sqrt(2)))<eps)
ck("r=3+2sqrt2: L = 2 ln(1+sqrt2): the square of door one, one doubling up",
    abs(R['L']-2*log(1+sqrt(2)))<eps)
ck("r=3+2sqrt2: theta = arccos(1/3) = 70.5288 deg", abs(cosh(R['L'])-3)<eps)
# ---- the inflection of G in L
Gfun = lambda L: 1 - L/(2*sinh(L/2))
Lstar = findroot(lambda L: diff(Gfun, L, 2), 3.2)
ck("inflection: L* = 3.212230597605..., r* = e^{L*} = 24.834...",
    abs(Lstar-mpf('3.212230597605'))<mpf('1e-11')
    and abs(exp(Lstar)-mpf('24.834'))<mpf('1e-3'))
ck("inflection: G' > 0 at every tested L (no extremum)", all(diff(Gfun,
    mpf(t)) > 0 for t in [0.1,0.5,1,2,3,5,8]))
# ---- Bernoulli radius
ck("Bernoulli series radius: |L/2| < pi <=> r < e^{2 pi} = 535.4916...",
    abs(exp(2*pi)-mpf('535.49165552476473'))<mpf('1e-10'))
# ---- the wall from the other side: the circular twin at the matched Cayley coordinate
tau = mpf(1)/3; beta = 2*atan(tau)
ck("twin at r=2's tau: beta = 2 arctan(1/3) = theta(2); beta/(2 sin(beta/2)) = "
    "1.01746459...",

    abs(beta-row(mpf(2))['theta'])<eps
    and abs(beta/(2*mp.sin(beta/2))-mpf('1.01746459032'))<mpf('1e-10'))
# ---- the offered row: 0(N) cells 2^{12/(N+8)}
for N, val in [(0, 2*sqrt(2)), (1, mpf(2)**(mpf(4)/3)), (2, mpf(2)**(mpf(6)/5)), (3,
    mpf(2)**(mpf(12)/11)), (4, mpf(2))]:
    ck(f"offered row: r_N = 2^{12/(N+8)} at N={N}", abs(mpf(2)**(mpf(12)/(N+8))-val)<eps)
ck("offered row: N = 4 lands on r = 2, i.e. c*a(4) = 2 ln2 * 3/12 = ln2/2",
    abs(2*ln2*mpf(3)/12-ln2/2)<eps)

```

```
# ---- the mirror: r and 1/r give the same G
ck("mirror:  $G(1/2) = G(2)$ ;  $G(1/(2+\sqrt{3})) = G(2+\sqrt{3})$ ",
    abs(row(mpf(1)/2)['G']-row(mpf(2))['G'])<eps
    and abs(row(1/(2+sqrt(3)))['G']-row(2+sqrt(3))['G'])<eps)
print(f"dps {dps:4d}    checks so far {CHECKS}    failures {FAILS}")

print("="*64); print(f"    CHECKS {CHECKS}    FAILURES {FAILS}"); print("="*64)
if FAILS: sys.exit(1)
```

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```
=====
CHECKS 176    FAILURES 0
=====
```

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Version 1.0, 2026-09. Rows from the thirteen papers as printed, every number in the block (44 checks at four precisions, 176 executions). Nothing in this note is a claim.